

Kartik Jain

Backend Engineer · Security Researcher

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Summary

Backend engineer and security researcher. I ship production REST API systems — payment, event, and media backends serving thousands of users — and report vulnerabilities to Meta, Microsoft, and CERT-In. Strong in Node.js, FastAPI, PostgreSQL, and application security.

Experience

R.U.D.R.A Cybersecurity — Security Engineer Intern, Mumbai May–Jul 2025

- Automated recurring external attack-surface monitoring with BBOT, reducing manual effort and saving approximately \$20,000/year.
- Single-handedly built a custom Semgrep (SAST) module with Neo4j-based visualization to improve large-scale code analysis.
- Discovered and reported 2 high-severity vulnerabilities (LFI, broken access control) in a banking client's web portal via manual code review; automated detection with dozens of custom Nuclei templates.
- Ran red-team / purple-team exercises with the Sliver C2 framework to emulate real-world intrusions.

Security Research

- Reported a privilege-escalation vulnerability in Meta's WhatsApp through its coordinated disclosure program.
- Reported access-control and information-disclosure flaws in Microsoft Intune and Exchange Admin to MSRC (VULN-150984).
- Responsibly disclosed a stored XSS on the Mumbai Police website in coordination with CERT-In.
- Reported critical exposures at The Souled Store (staging infrastructure) and Belgian Waffle Co. (source-code and credential leak).
- Defensive: hardened infrastructure protecting 16,000+ users' data and led incident response to contain a ransomware attack via rapid backup-and-recovery.

Projects

- GhostShare — end-to-end-encrypted, zero-knowledge file sharing. Keys are derived in-browser with Argon2id so the server only ever stores ciphertext, with realtime sync and chunked direct-to-Cloudflare R2 uploads. TypeScript, Next.js, Fastify, PostgreSQL, Redis. [\[Live\]](#)
- bad-db — a database storage engine written from scratch in Java: a custom on-disk format, a hand-rolled B-Tree index, tombstone deletes with compaction, and its own SQL-like query language (tokenizer, parser, and REPL). [\[Source\]](#)
- Event App Backends — lead author of Python (FastAPI / Flask) backends for NMIMS's largest events: the Taqneeq credits/loyalty system (OTP-based authentication, referrals, cached live leaderboard) and the Mumbai MUN (Delego) backend, which shipped to the App Store for 600+ delegates. [\[Taqneeq App\]](#) [\[Source\]](#)
- Event Registration Platform — full-stack paid registration with Cashfree payments, HMAC-verified webhooks, QR ticket validation, and an admin analytics dashboard, plus role-scoped authorization (RBAC). [\[Source\]](#)

Achievements

- 1st place (global leaderboard), AI Jailbreakathon by chat.win — LLM prompt attack & defense, won with the fewest tokens.
- 1st place, CodeForge 2026 hackathon (Taqneeq & FOSS MPSTME) — 3-person team; NIFTY options-analytics platform.

Skills

Languages: Python, TypeScript, JavaScript, Java, C++, SQL

Backend & DevOps: Node.js, Express, FastAPI, Flask, REST APIs, JWT, PostgreSQL, Redis, Drizzle ORM, MongoDB, Docker, Nginx, CI/CD, AWS, GCP, Linux

Security: Penetration testing, vulnerability assessment, SAST/DAST, OWASP Top 10, Burp Suite, Nmap, Semgrep, Nuclei, BBOT, Sliver C2, Metasploit, Wireshark, OWASP ZAP, Neo4j

Education & Leadership

- B.Tech, Information Technology — NMIMS MPSTME, Mumbai 2024–2028
- Chairperson, [ACM MPSTME](#) (Jun 2026–present); previously Head of Technical Department at Taqneeq, Mumbai MUN, Social Conclave, and Sattva (2025–26).